

GEN AI LAB – 2

Name - Ayush Pathak

SRN – PES2UG23CS113

1. The Power of Framework Abstraction (LangChain)

The transition from raw API calls to using LangChain highlighted the importance of "Composability" in AI development. By using the Pipe Operator (|) and LCEL (LangChain Expression Language), I observed that I could build complex logic pipelines that are far easier to read and maintain than manual "spaghetti code". The ability to swap a model or add an output parser by changing just one line of code demonstrates how a framework can future proof an application.

2. Controlling Model "Brain" Configuration

I observed that an LLM is not a static tool but a configurable engine. The Temperature parameter proved to be the most critical "knob" for controlling output:

- Precision (Temp=0): Essential for tasks requiring consistency, such as the "Accountant" model or our MoE Router, where the same input must always lead to the same classification.
- Creativity (Temp=1): Better for brainstorming or creative writing, where diversity in responses is valued.

3. Prompt Engineering as Probability Manipulation

One of the most profound realizations was that LLMs are Stochastic (Random) next token predictors. Prompt Engineering is essentially the art of manipulating these probabilities. I learned that Ambiguity is the Enemy; every detail left out of a prompt is a gap the model must fill with statistical guesswork, often leading to hallucinations. By using the CO STAR Framework, I could provide the necessary context, constraints, and style to force the model into a high precision state.

4. Advanced Reasoning: CoT, ToT, and GoT

I observed that while standard models try to jump straight to an answer, forcing them to "think out loud" via Chain of Thought (CoT) significantly improves logic. This "Latent Reasoning" allows the model to attend to its own intermediate steps to solve multi step problems. Furthermore:

- Tree of Thoughts (ToT): Showed how a "Judge" can evaluate multiple parallel branches to find the best solution.
- Graph of Thoughts (GoT): Demonstrated how information can be split and then Aggregated back together, mimicking a "Writer's Room" workflow.

5. Architecting Scalable Intelligence (Mixture of Experts)

The Mixture of Experts (MoE) assignment was a practical application of routing logic. I learned that a "Generalist" AI is often less effective than a "Specialist". By building a Router to classify intent and an Orchestrator to select the correct specialized system prompt, I created a system that could handle Technical, Billing, and General queries with domain specific precision.

6. Solving the "Hallucination" Problem with RAG

Finally, I explored Retrieval Augmented Generation (RAG) as an "Open Book Test" for AI. I observed that converting text into Embeddings (Vectors) allows a computer to understand semantic similarity mathematically through Cosine Similarity. Lessons on indexing algorithms like FAISS, IVF, and HNSW highlighted the engineering trade offs between speed, accuracy, and memory when searching through millions of documents.